

Module 01 Assignments

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Group 7.1

Question 5

$$\int t e^{2t} dt$$

let

$$u = t \quad dv = e^{2t} dt$$

then

$$du = dt \quad v = \frac{1}{2} e^{2t}$$

substitute into IBP formula

$$\int u dv = uv - \int v du$$

$$\int t e^{2t} dt = t \frac{1}{2} e^{2t} - \int \frac{1}{2} e^{2t} dt$$

integrate the remaining term

$$\int \frac{1}{2} e^{2t} dt = \frac{1}{2} \left(\frac{1}{2} e^{2t} \right) = \frac{1}{4} e^{2t}$$

$$\boxed{\frac{1}{2} t e^{2t} - \frac{1}{4} e^{2t} + C}$$

Question 8

$$\int (\pi - x) \cos(\pi x) dx$$

let

$$u = \pi - x \quad dv = \cos(\pi x) dx$$

then

$$du = -dx \quad v = \frac{1}{\pi} \sin(\pi x)$$

substitute into IBP formula

$$\int u dv = uv - \int v du$$

$$\int (\pi - x) \cos(\pi x) dx = (\pi - x) \frac{1}{\pi} \sin(\pi x) - \int \frac{1}{\pi} \sin(\pi x) (-dx)$$

integrate the remaining term

$$- \int \frac{1}{\pi} \sin(\pi x) (dx) = - \left[\frac{1}{\pi} \left(-\frac{1}{\pi} \cos(\pi x) \right) \right] = \frac{1}{\pi^2} \cos(\pi x)$$

$$\frac{\pi - x}{\pi} \sin(\pi x) - \frac{1}{\pi^2} \cos(\pi x) + C$$

Question 10

$$\int \frac{\ln x}{x^2} dx$$

let

$$u = \ln x \quad dv = x^{-2} dx$$

then

$$du = \frac{1}{x} dx \quad v = -x^{-1}$$

substitute into IBP formula

$$\int u \, dv = uv - \int v \, du$$

$$\int \frac{\ln x}{x^2} \, dx = -\frac{\ln x}{x} - \int -x^{-1} \frac{1}{x} \, dx$$

integrate the remaining term

$$- \int \left(-\frac{1}{x}\right) \left(\frac{1}{x}\right) \, dx = \int \frac{1}{x^2} \, dx = \int x^{-2} \, dx = -x^{-1}$$

$$\boxed{-\frac{\ln x}{x} - \frac{1}{x} + C}$$

Question 12

$$\int t^2 \sin(\beta t) \, dt$$

let

$$u = t^2 \quad dv = \sin(\beta t) \, dt$$

then

$$du = 2t \, dt \quad v = -\frac{1}{\beta} \cos(\beta t)$$

substitute into IBP formula

$$\int u \, dv = uv - \int v \, du$$

$$\int t^2 \sin(\beta t) \, dt = t^2 \left(-\frac{1}{\beta} \cos(\beta t)\right) - \int \left(-\frac{1}{\beta} \cos(\beta t)\right) (2t) \, dt$$

factor the constants

$$= -\frac{t^2}{\beta} \cos(\beta t) + \frac{2}{\beta} \int t \cos(\beta t) \, dt$$

The remaining integral still contains a product of:

$$t \rightarrow P \quad \cos(\beta t) \rightarrow T$$

apply the IBP formula again

$$u = t \quad dv = \cos(\beta t) dt$$

then

$$du = dt \quad v = \frac{1}{\beta} \sin(\beta t)$$

so

$$\int t \cos(\beta t) dt = t \left(\frac{1}{\beta} \sin(\beta t) \right) - \int \frac{1}{\beta} \sin(\beta t) dt$$

solve the remaining integral

$$\int \frac{1}{\beta} \sin(\beta t) dt = \frac{1}{\beta} \left(-\frac{1}{\beta} \cos(\beta t) \right) = -\frac{1}{\beta^2} \cos(\beta t)$$

therefore

$$\frac{2}{\beta} \int t \cos(\beta t) dt = \frac{2}{\beta} \left[t \left(\frac{1}{\beta} \sin(\beta t) \right) - \frac{1}{\beta^2} \cos(\beta t) \right]$$

replace in the original IBP formula

$$-\frac{t^2}{\beta} \cos(\beta t) + \frac{2t}{\beta^2} \sin(\beta t) + \frac{2}{\beta^3} \cos(\beta t) + C$$

Question 14

$$\int \ln(\sqrt{x}) dx$$

let

$$u = \ln(\sqrt{x}) \quad dv = dx$$

then

$$du = \frac{1}{2x} dx \quad v = x$$

substitute into IBP formula

$$\int u dv = uv - \int v du$$

$$\int \ln(\sqrt{x}) \, dx = x \ln(\sqrt{x}) - \int \frac{x}{2x} dx$$

integrate the remaining term

$$- \int \frac{x}{2x} dx = \frac{1}{2} dx = \frac{x}{2}$$

$$\boxed{x \ln(\sqrt{x}) - \frac{x}{2} + C}$$

Question 16

$$\int \tan^{-1}(2y) \, dy$$

let

$$u = \tan^{-1}(2y) \quad dv = dy$$

then

$$du = \frac{2}{1+4y^2} dy \quad v = y$$

substitute into IBP formula

$$\int u \, dv = uv - \int v \, du$$

$$\int \tan^{-1}(2y) \, dy = y \tan^{-1}(2y) - \int \frac{2y}{1+4y^2} dy$$

for the remaining integral, use substitution

$$u = 1 + 4y^2$$

then

$$du = 8y \, dy$$

and therefore

$$2y \, dy = \frac{1}{4} du$$

substitute:

$$\int \frac{2y}{1+4y^2} dy = \frac{1}{4} \int \frac{1}{u} du$$

apply the logarithmic rule and substitute

$$\frac{1}{4} \ln |1+4y^2|$$

Therefore,

$$\boxed{y \tan^{-1}(2y) - \frac{1}{4} \ln(1+4y^2) + C}$$

Question 19

$$\int (\ln x)^2 dx$$

let

$$u = (\ln x)^2 \quad dv = dx$$

then

$$du = \frac{2 \ln x}{x} dx \quad v = x$$

substitute into IBP formula

$$\int (\ln x)^2 dx = x(\ln x)^2 - \int x \left(\frac{2 \ln x}{x} \right) dx$$

the remaining integral still requires integration by parts

$$2 \int \ln x dx$$

let

$$u = \ln x \quad dv = dx$$

then

$$du = \frac{1}{x} \quad v = x$$

apply IBP again

$$2 \int \ln x \, dx = x \ln x - \int \frac{x}{x} \, dx$$

therefore

$$\boxed{\int (\ln x)^2 \, dx = x(\ln x)^2 - 2x \ln x + 2x + C}$$

Question 24

$$\int e^{-\theta} \cos(2\theta) \, d\theta$$

First Integration by Parts

$$u = e^{-\theta} \quad dv = \cos(2\theta) \, d\theta$$

then

$$du = -e^{-\theta} \, d\theta \quad v = \frac{1}{2} \sin(2\theta)$$

substitute into IBP formula

$$\int e^{-\theta} \cos(2\theta) \, d\theta = \frac{1}{2} e^{-\theta} \sin(2\theta) + \frac{1}{2} \int e^{-\theta} \sin(2\theta) \, d\theta$$

Second Integration by Parts

For

$$\int e^{-\theta} \sin(2\theta) \, d\theta$$

choose

$$u = e^{-\theta} \quad dv = \sin(2\theta) \, d\theta$$

then

$$du = -e^{-\theta} \, d\theta \quad v = -\frac{1}{2} \cos(2\theta)$$

so

$$\int e^{-\theta} \sin(2\theta) \, d\theta = -\frac{1}{2} e^{-\theta} \cos(2\theta) - \frac{1}{2} \int e^{-\theta} \cos(2\theta) \, d\theta$$

The original integral appears as the last term on the right side of the equation.

$$\int e^{-\theta} \cos(2\theta) d\theta = \frac{1}{2} e^{-\theta} \sin(2\theta) + \frac{1}{2} \left[-\frac{1}{2} e^{-\theta} \cos(2\theta) - \frac{1}{2} \int e^{-\theta} \cos(2\theta) d\theta \right]$$

Distribute $\frac{1}{2}$

$$\frac{1}{2} \left[-\frac{1}{2} e^{-\theta} \cos(2\theta) - \frac{1}{2} \int e^{-\theta} \cos(2\theta) d\theta \right] = -\frac{1}{4} e^{-\theta} \cos(2\theta) - \frac{1}{4} \int e^{-\theta} \cos(2\theta) d\theta$$

Add $\frac{1}{4} \int e^{-\theta} \cos(2\theta) d\theta$ to both sides.

$$\frac{5}{4} \int e^{-\theta} \cos(2\theta) d\theta = \frac{1}{2} e^{-\theta} \sin(2\theta) - \frac{1}{4} e^{-\theta} \cos(2\theta)$$

Multiply both sides by $\frac{4}{5}$:

$$\int e^{-\theta} \cos(2\theta) d\theta = \frac{2}{5} e^{-\theta} \sin(2\theta) - \frac{1}{5} e^{-\theta} \cos(2\theta)$$

Therefore,

$$\boxed{\int e^{-\theta} \cos(2\theta) d\theta = \frac{e^{-\theta}}{5} (2\sin(2\theta) - \cos(2\theta)) + C}$$

Question 48

$$\int \frac{\arcsin(\ln x)}{x} dx$$

Substitution followed by Integration by Parts

First: use substitution

Let

$$u = \ln x$$

then

$$du = \frac{1}{x} dx$$

so the integral becomes

$$\int \arcsin(u) du$$

Apply the Integration by Parts

Choose

$$w = \arcsin(u) \quad dv = du$$

then

$$dw = \frac{1}{\sqrt{1-u^2}} du \quad v = u$$

Apply the IBP formula:

$$\int \arcsin(u) du = u \arcsin(u) - \int u \left(\frac{1}{\sqrt{1-u^2}} \right) du$$

Solve the remaining integral with substitution

Let

$$z = 1 - u^2$$

then

$$dz = -2u du$$

so

$$u du = -\frac{1}{2} dz$$

Therefore,

$$\int \frac{u}{\sqrt{1-u^2}} du = -\frac{1}{2} \int z^{-\frac{1}{2}} dz$$

Apply the power rule

$$-\frac{1}{2} (2z^{\frac{1}{2}}) = -\sqrt{z}$$

Substitute back:

$$-\sqrt{1-u^2}$$

so

$$u \arcsin(u) + \sqrt{1-u^2} + C$$

Substitute $u = \ln x$

$$\boxed{(\ln x) \arcsin(\ln x) + \sqrt{1 - (\ln x)^2} + C}$$

Group 7.2

Question 1

Evaluate the integral

$$\int \sin^3(x) \cos^2(x) dx$$

Since the power of sin is odd, separate one factor of $\sin x$.

$$\sin^3(x) = \sin^2(x) \sin(x)$$

So

$$\int \sin^3(x) \cos^2(x) dx = \int \sin^2(x) \sin(x) \cos^2(x) dx$$

Rewrite the integral using the Pythagorean identity $\sin^2 x = 1 - \cos^2 x$.

$$\int (1 - \cos^2(x)) \cos^2(x) \sin(x) dx$$

Recognize that $-\sin(x)$ is the derivative of $\cos(x)$, so let $u = \cos(x)$.

$$u = \cos(x) \quad du = -\sin(x) dx$$

Apply the u-substitution to rewrite the integral.

$$-\int (1 - u^2) u^2 du$$

Distribute u^2 across the parentheses.

$$-\int (u^2 - u^4) du$$

Find the antiderivative of the expression in terms of u .

$$\frac{u^5}{5} - \frac{u^3}{3} + C$$

Rewrite in terms of x .

$$\boxed{\frac{1}{5} \cos^5(x) - \frac{1}{3} \cos^3(x) + C}$$

Question 2

Evaluate the integral

$$\int \cos^6(y) \sin^3(y) dy$$

Since the power of $\sin$ is odd, separate one factor of $\sin(y)$.

$$\sin^3(y) = \sin^2(y) \sin(y)$$

So

$$\int \cos^6(y) \sin^3(y) dy = \int \cos^6(y) \sin^2(y) \sin(y) dy$$

Rewrite the integral using the Pythagorean identity $\sin^2(y) = 1 - \cos^2(y)$.

$$\int \cos^6(y) (1 - \cos^2(y)) \sin(y) dy$$

Recognize that $-\sin(y)$ is the derivative of $\cos(y)$, so let $u = \cos(y)$.

$$u = \cos(y) \quad du = -\sin(y) dy$$

Apply the u -substitution to rewrite the integral.

$$-\int u^6 (1 - u^2) du$$

Distribute u^6 across the parentheses.

$$-\int (u^6 - u^8) du$$

Find the antiderivative of the expression in terms of u .

$$\frac{u^9}{9} - \frac{u^7}{7} + C$$

Rewrite in terms of y .

$$\boxed{\frac{1}{9} \cos^9(y) - \frac{1}{7} \cos^7(y) + C}$$

Question 4

Evaluate the integral

$$\int_0^{\frac{\pi}{4}} \sin^5(x) dx$$

Since the power of $\sin$ is odd, separate one factor of $\sin(x)$.

$$\int_0^{\frac{\pi}{4}} (\sin^2(x))^2 \sin(x) dx$$

Rewrite the integral using the Pythagorean identity $\sin^2(x) = 1 - \cos^2(x)$.

$$\int_0^{\frac{\pi}{4}} (1 - \cos^2(x))^2 \sin(x) dx$$

Recognize that $-\sin(x)$ is the derivative of $\cos(x)$, so let $u = \cos(x)$.

$$u = \cos(x) \quad du = -\sin(x) dx$$

So

$$\sin(x) dx = -du$$

Convert the bounds to u -values.

When

$$x = 0$$

$$u = \cos(0) = 1$$

When

$$x = \frac{\pi}{4}$$

$$u = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

Apply the u -substitution to rewrite the integral

$$- \int_1^{\frac{\sqrt{2}}{2}} (1 - u^2)^2 du$$

Reverse the bounds to remove the negative.

$$\int_{\frac{\sqrt{2}}{2}}^1 (1 - u^2)^2 du$$

Expand the squared binomial.

$$\int_{\frac{\sqrt{2}}{2}}^1 (1 - 2u^2 + u^4)^2 du$$

Find the antiderivative of the expression in terms of u .

$$\left[u - \frac{2}{3}u^3 + \frac{1}{5}u^5 \right]_{\frac{\sqrt{2}}{2}}^1$$

Plug in $u=1$

$$1 - \frac{2}{3} + \frac{1}{5} = \frac{8}{15}$$

The plug in $u = \frac{\sqrt{2}}{2}$.

$$\frac{\sqrt{2}}{2} - \frac{2}{3} \left(\frac{\sqrt{2}}{2} \right)^3 + \frac{1}{5} \left(\frac{\sqrt{2}}{2} \right)^5$$

Simplify the powers and coefficients.

$$= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{6} + \frac{\sqrt{2}}{40}$$

Combine using the LCD 120.

$$\frac{60\sqrt{2} - 20\sqrt{2} + 3\sqrt{2}}{120} = \frac{43\sqrt{2}}{120}$$

Evaluate the upper and lower bound.

$$\frac{8}{15} - \frac{43\sqrt{2}}{120} = \frac{64 - 43\sqrt{2}}{120}$$

$$\boxed{\frac{64 - 43\sqrt{2}}{120}}$$

Question 7

Evaluate the integral

$$\int_0^{\frac{\pi}{2}} \cos^2(\theta) d\theta$$

Recognize the half angle identity for $\cos^2(\theta)$.

$$\cos^2(\theta) = \frac{1}{2}(1 + \cos(2\theta))$$

Rewrite the integral using the half angle identity.

$$\int_0^{\frac{\pi}{2}} \cos^2(\theta) d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 + \cos(2\theta) d\theta$$

Find the antiderivative of the integral.

$$\left[\frac{1}{2} \left(\theta + \frac{1}{2} \sin(2\theta) \right) \right]_0^{\frac{\pi}{2}}$$

Evaluate at the bounds.

$$\frac{1}{2} \left(\frac{\pi}{2} + \frac{1}{2} \sin(2 \cdot \frac{\pi}{2}) \right) - \frac{1}{2} \left(0 + \frac{1}{2} \sin(2 \cdot 0) \right)$$

Then

$$\sin(\pi) = 0 \quad \sin(0) = 0$$

so this becomes

$$\frac{1}{2} \left(\frac{\pi}{2} \right) - 0 = \frac{\pi}{4}$$

$$\boxed{\frac{\pi}{4}}$$

Question 16

Evaluate the integral

$$\int \csc^5(\theta) \cos^3(\theta) d\theta$$

Recognize the reciprocal identity $\csc(\theta) = \frac{1}{\sin(\theta)}$ and rewrite the expression in terms of sine and cosine.

$$\int \csc^5(\theta) \cos^3(\theta) d\theta = \frac{\cos^3(\theta)}{\sin^5(\theta)}$$

Since the power of $\cos$ is odd, separate one factor of $\cos(\theta)$.

$$\frac{\cos^2(\theta) \cos(\theta)}{\sin^5(\theta)}$$

Rewrite the integral using the Pythagorean identity $\cos^2(\theta) = 1 - \sin^2(\theta)$.

$$\int \frac{1 - \sin^2(\theta) \cos(\theta)}{\sin^5(\theta)} d\theta$$

Recognize that $\cos(\theta)$ is the derivative of $\sin(\theta)$, so let $u = \sin(\theta)$.

$$u = \sin(\theta) \quad du = \cos(\theta) d\theta$$

Apply the u -substitution to rewrite the integrand

$$\frac{1 - u^2}{u^5}$$

Rewrite as a product.

$$(1 - u^2)u^{-5}$$

Distribute u^{-5} .

$$u^{-5} - u^{-3}$$

Find the antiderivative of the expression in terms of u .

$$\frac{u^{-2}}{2} - \frac{u^{-4}}{4} + C$$

Rewrite in terms of θ .

$$\boxed{\frac{1}{2} \sin^2(\theta) - \frac{1}{4} \sin^4(\theta) + C}$$

Question 21

Evaluate the integral

$$\int \tan(x) \sec^3(x) dx$$

Recognize that $\sec(x)\tan(x)$ is the derivative of $\sec(x)$, so let $u = \sec(x)$.

$$u = \sec(x) \quad du = \sec(x)\tan(x) dx$$

Apply the u -substitution to rewrite the integrand

$$u^2$$

Find the antiderivative of the expression in terms of u .

$$\frac{u^3}{3} + C$$

Rewrite in terms of x .

$$\boxed{\frac{1}{3} \sec^3(x) + C}$$

Question 23

Evaluate the integral

$$\int \tan^2(x) dx$$

Use the identity

$$\tan^2(x) = \sec^2(x) - 1$$

So

$$\int \tan^2(x) dx = \int (\sec^2(x) - 1) dx$$

Integrate

$$\boxed{\tan(x) - x + C}$$

Question 26

Evaluate the integral

$$\int_0^{\frac{\pi}{4}} \sec^6(\theta) \tan^6(\theta) d\theta$$

Factor $\sec^2(x)$ from the original integral.

$$\int_0^{\frac{\pi}{4}} (\sec^2(\theta))^2 \sec^2(\theta) \tan^6(\theta) d\theta$$

Rewrite the integrand using the Pythagorean identity.

$$\sec^2(\theta) = 1 + \tan^2(\theta)$$

So

$$(1 + \tan^2(\theta))^2 \sec^2(\theta) \tan^6(\theta)$$

Recognize that $\sec^2(\theta)$ is the derivative of $\tan(\theta)$, so let $u = \sec^2(\theta)$.

$$u = \tan(\theta) \quad du = \sec^2(\theta) d\theta$$

Convert the bounds to u -values.

At $\theta = 0$

$$\tan(0) = 0$$

At $\theta = \frac{\pi}{4}$

$$\tan\left(\frac{\pi}{4}\right) = 1$$

Apply the u -substitution to rewrite the integral.

$$\int_0^1 (1 + u^2)^2 u^6 du$$

Expand the binomial and distribute u^6 .

$$u^6 + 2u^8 + u^{10}$$

Find the antiderivative of the expression in terms of u .

$$\frac{u^7}{7} + 2\frac{u^9}{9} + \frac{u^{11}}{11} + C$$

Evaluate at the bounds.

$$\left(\frac{1^7}{7} + 2\frac{1^9}{9} + \frac{1^{11}}{11}\right) - \left(\frac{0^7}{7} + 2\frac{0^9}{9} + \frac{0^{11}}{11}\right)$$

so this becomes

$$\frac{99 + 154 + 63}{693} - 0 = \frac{316}{693}$$

$$\frac{316}{693}$$

Question 28

Evaluate the integral

$$\int \tan^5(x) \sec^3(x) dx$$

Because the power of tangent is odd, save a factor of $\sec(x) \tan(x)$

Rewrite

$$\tan^4(x) \sec^2(x) (\sec(x) \tan(x))$$

Use the Pythagorean identity and substitute in the integrand

$$(\sec^2(x) - 1)^2 \sec^2(x) (\sec(x) \tan(x))$$

Let

$$u = \sec(x) \quad du = \sec(x) \tan(x) dx$$

Rewrite in terms of u

$$\int (u^2 - 1)^2 u^2 \, du$$

Expand and simplify

$$\int u^6 - 2u^4 + u^2 \, du$$

Integrate

$$\frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3} + C$$

Substitute back $u = \sec(x)$

$$\boxed{\frac{1}{7} \sec^7(x) - \frac{2}{5} \sec^5(x) + \frac{1}{3} \sec^3(x) + C}$$

Group 7.3

Question 3

$$\frac{x^2}{\sqrt{x^2 - 2}} \, dx$$

a. Determine an appropriate trigonometric substitution.

$$x = \sqrt{2} \sec(\theta)$$

b. Apply the substitution to transform the integral into a trigonometric integral. Do not evaluate the integral.

Find dx

$$dx = \sqrt{2} \sec(\theta) \tan(\theta) \, d\theta$$

Simplify the radical

$$\sqrt{x^2 - 2} = \sqrt{2 \sec^2(\theta) - 2}$$

$$= \sqrt{2(\sec^2(\theta) - 1)}$$

Recognize the Pythagorean identity $\sec^2(\theta) - 1 = \tan^2(\theta)$

$$\sqrt{2 \tan^2(\theta)} = \sqrt{2} \tan(\theta)$$

Transform the integral

$$\int \frac{2 \sec^2(\theta)}{\sqrt{2} \tan(\theta)} (\sec(\theta) \sqrt{2} \tan(\theta)) d\theta$$

Cancel common factors

$$\boxed{2 \int \sec^3(\theta) d\theta}$$

Question 6

Evaluate the integral using the indicated trigonometric substitution. Sketch and label the associated right triangle.

$$\int \frac{x^3}{\sqrt{9+x^2}} dx \quad x = 3 \tan(\theta)$$

Differentiate the substitution

$$dx = 3 \sec^2(\theta) d\theta$$

Simplify the radical

$$\sqrt{9+x^2} = \sqrt{9+9 \tan^2(\theta)}$$

$$= \sqrt{9(1+\tan^2(\theta))}$$

Recognize the Pythagorean identity $1 + \tan^2(\theta) = \sec^2(\theta)$

$$\sqrt{9 \sec^2(\theta)} = 3 \sec(\theta)$$

Substitute and simplify

$$\int \frac{(3 \tan(\theta))^3}{3 \sec(\theta)} (3 \sec^2(\theta)) d\theta = 27 \int \tan^3(\theta) \sec(\theta) d\theta$$

Since the power of $\tan$ is odd, separate one factor of $\tan(\theta)$.

$$\tan^3(\theta) = \tan^2(\theta) \tan(\theta)$$

So

$$27 \int \tan^2(\theta) \tan(\theta) \sec(\theta) d\theta$$

Rewrite the integral using the Pythagorean identity $\tan^2(\theta) = \sec^2(\theta) - 1$.

$$(\sec^2(\theta) - 1) \sec(\theta) \tan(\theta)$$

Apply u -substitution

$$u = \sec(\theta) \quad du = \sec(\theta) \tan(\theta) d\theta$$

Rewrite in terms of u

$$27 \int (u^2 - 1) du$$

Find the antiderivative

$$27 \left(\frac{u^3}{3} - u \right) + C$$

Rewrite in terms of $\sec(\theta)$

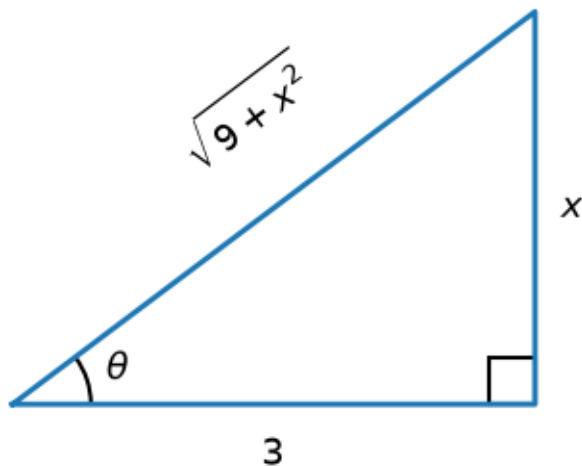
$$9 \sec^3(\theta) - 27 \sec(\theta) + C$$

Sketch and label the associated right triangle

$$x = 3 \tan(\theta) \quad \tan(\theta) = \frac{x}{3} = \frac{\textit{opposite}}{\textit{adjacent}}$$

By the Pythagorean theorem

$$\text{hypotenuse} = \sqrt{9 + x^2}$$



From the triangle

$$\sec(\theta) = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{\sqrt{9+x^2}}{3}$$

Rewrite in terms of x

$$9 \left(\frac{\sqrt{9+x^2}}{3} \right)^3 - 27 \left(\frac{\sqrt{9+x^2}}{3} \right) + C$$

Simplify

$$\frac{9}{27} (9+x^2)^{\frac{3}{2}} - 9\sqrt{9+x^2} + C$$

$$\boxed{\sqrt{9+x^2} \left(\frac{1}{3}x^2 - 6 \right) + C}$$

Question 10

Evaluate the integral.

$$\int \frac{x^2}{\sqrt{9-x^2}} dx$$

Choose the trigonometric substitution

$$x = 3 \sin(\theta) \quad dx = 3 \cos(\theta) d\theta$$

Simplify the radical

$$\begin{aligned}\sqrt{9 - x^2} &= \sqrt{9 - 9\sin^2(\theta)} \\ &= 3\sqrt{1 - \sin^2(\theta)}\end{aligned}$$

Recognize the Pythagorean identity $1 - \sin^2(\theta) = \cos^2(\theta)$

$$3\sqrt{\cos^2(\theta)} = 3 \cos(\theta)$$

Apply the trigonometric substitution

$$\int \frac{9 \sin^2(\theta)}{3 \cos(\theta)} 3 \cos(\theta) d\theta = 9 \int \sin^2(\theta) d\theta$$

Apply the half-angle identity

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

Therefore

$$\frac{9}{2} \int (1 - \cos(2\theta)) d\theta$$

Integrate

$$= \frac{9}{2} \left(\theta - \frac{1}{2} \sin(2\theta) \right) + C$$

Back-substitute

From

$$x = 3 \sin(\theta)$$

we have

$$\sin(\theta) = \frac{x}{3}, \quad \theta = \arcsin\left(\frac{x}{3}\right)$$

Using the associated right triangle

$$\cos(\theta) = \frac{\sqrt{9-x^2}}{3}$$

Convert the last term in the antiderivative using the double angle identity.

$$\frac{1}{2} \sin(2\theta) = \frac{1}{2} (2 \sin(\theta) \cos(\theta))$$

Rewrite in terms of x

$$\left(\frac{x}{3}\right) \left(\frac{\sqrt{9-x^2}}{3}\right) = \frac{x\sqrt{9-x^2}}{9}$$

Substitute back

$$\frac{9}{2} \left[\arcsin\left(\frac{x}{3}\right) - \frac{x\sqrt{9-x^2}}{9} \right] + C$$

Simplify

$$\boxed{\frac{9}{2} \arcsin\left(\frac{x}{3}\right) - \frac{x\sqrt{9-x^2}}{2} + C}$$

Question 12

Evaluate the integral.

$$\int_0^3 \frac{x}{\sqrt{36-x^2}} dx$$

Choose the trigonometric substitution

$$x = 6 \sin(\theta) \quad dx = 6 \cos(\theta) d\theta$$

Simplify the radical

$$\sqrt{36-x^2} = \sqrt{36-36\sin^2(\theta)}$$

$$6\sqrt{1-\sin^2(\theta)} = 6\cos(\theta)$$

Substitute

$$\int \frac{6 \sin(\theta)}{6 \cos(\theta)} 6 \cos(\theta) d\theta$$

which simplifies to

$$\int 6 \sin(\theta) d\theta$$

Change the bounds

For $x = 0$

$$0 = 6 \sin(\theta) \Rightarrow \theta = 0$$

For $x = 3$

$$3 = 6 \sin(\theta) \Rightarrow \theta = \frac{\pi}{6}$$

Therefore

$$\int_0^{\frac{\pi}{6}} 6 \sin(\theta) d\theta$$

Find the antiderivative

$$[-6 \cos(\theta)]_0^{\frac{\pi}{6}}$$

Evaluate at the bounds

$$-6 \cos\left(\frac{\pi}{6}\right) - (-6 \cos(0))$$

$$\boxed{6 - 3\sqrt{3}}$$

Question 15

Evaluate the integral.

$$\int_2^3 \frac{dx}{(x^2 - 1)^{\frac{3}{2}}}$$

Choose the trigonometric substitution

$$x = \sec(\theta) \quad dx = \sec(\theta)\tan(\theta) d\theta$$

Simplify the radical

$$(x^2 - 1)^{\frac{3}{2}} = (\sec^2(\theta) - 1)^{\frac{3}{2}}$$

So

$$\tan^3(\theta)$$

Substitute and simplify

$$\int \frac{\sec(\theta) \tan(\theta)}{\tan^3(\theta)} d\theta = \int \frac{\sec(\theta)}{\tan^2(\theta)} d\theta$$

Rewrite using sine and cosine identities

$$\frac{\sec(\theta)}{\tan^2(\theta)} = \frac{\cos(\theta)}{\sin^2(\theta)}$$

Recognize reciprocal and quotient identities

$$\frac{1}{\sin(\theta)} \frac{\cos(\theta)}{\sin(\theta)} = \csc(\theta) \cot(\theta)$$

Therefore

$$\int \csc(\theta) \cot(\theta) d\theta$$

Convert the bounds in terms of θ

For $x = 3$

$$x = \sec(\theta) = 3 \Rightarrow \cos(\theta) = \frac{1}{3} \Rightarrow \theta = \arccos(1/3)$$

For $x = 2$

$$x = \sec(\theta) = 2 \Rightarrow \cos(\theta) = \frac{1}{2} \Rightarrow \theta = \arccos(1/2)$$

So

$$\int_{\frac{\pi}{3}}^{\arccos(\frac{1}{3})} \csc(\theta) \cot(\theta) d\theta$$

Integrate

$$- [\csc(\theta)]_{\frac{\pi}{3}}^{\arccos(\frac{1}{3})}$$

Evaluate the special angle bound

For

$$\csc\left(\frac{\pi}{3}\right)$$

use the unit-circle value

$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

Therefore

$$\csc\left(\frac{\pi}{3}\right) = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

Evaluate the inverse-trig bound with a right triangle

Now consider

$$\csc\left(\arccos\left(\frac{1}{3}\right)\right)$$

Let

$$\theta = \arccos\left(\frac{1}{3}\right)$$

Then

$$\cos(\theta) = \frac{1}{3} = \frac{\text{adjacent}}{\text{hypotenuse}}$$

Find the opposite side

$$\text{opposite} = \sqrt{3^2 - 1^2} = \sqrt{8}$$

Therefore

$$\sin(\theta) = \frac{2\sqrt{2}}{3}$$

and hence

$$\csc(\theta) = \frac{3}{2\sqrt{2}} = \frac{6\sqrt{2}}{8}$$

Place the evaluated values

$$- \left[\frac{2}{3}\sqrt{3} - \frac{3}{4}\sqrt{2} \right]$$

Simplify using the LCD and distribute the negative

$$\boxed{\frac{9\sqrt{2} - 8\sqrt{3}}{12}}$$

Question 18

Evaluate the integral.

$$\int_0^2 \frac{dt}{\sqrt{4+t^2}}$$

Find the trigonometric substitution and simplify the radical

$$t = 2 \tan(\theta) \quad dt = 2 \sec^2(\theta) d\theta$$

$$\sqrt{4 + 4 \tan^2(\theta)} = 2 \sec(\theta)$$

Rewrite the integral and integrate

$$\int \frac{2 \sec^2(\theta)}{2 \sec(\theta)} d\theta = \int \sec(\theta) d\theta = \ln |\sec(\theta) + \tan(\theta)| + C$$

Convert the bounds $t = 2 \tan(\theta)$

At $t = 0$

$$0 = 2 \tan(\theta) \Rightarrow \tan(\theta) = 0 \Rightarrow \theta = 0$$

At $t = 2$

$$2 = 2 \tan(\theta) \Rightarrow \tan(\theta) = 1$$

Sine and cosine are equal at $\theta = \frac{\pi}{4}$

Evaluate the antiderivative at the new bounds

$$[\ln |\sec(\theta) + \tan(\theta)|]_0^{\frac{\pi}{4}}$$

Upperbound

$$\sec\left(\frac{\pi}{4}\right) = \frac{1}{\cos\left(\frac{\pi}{4}\right)} = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2}$$

$$\tan\left(\frac{\pi}{4}\right) = 1$$

$$\ln |\sqrt{2} + 1|$$

Lowerbound

$$\sec(0) = \frac{1}{\cos(0)} = 1$$

$$\tan(0) = \frac{\sin(0)}{\cos(0)} = 0$$

$$\ln |1| = 0$$

Therefore

$$\boxed{\ln |\sqrt{2} + 1|}$$

Question 21

Evaluate the integral.

$$\int_0^a x^2 \sqrt{a^2 - x^2} \, dx$$

Substitute the trigonometric expression and simplify the radical

$$x = a \sin(\theta) \quad dx = a \cos(\theta) d\theta$$

$$\sqrt{a^2 - a^2 \sin^2(\theta)} = a \cos(\theta)$$

Rewrite in terms of θ and simplify

$$(a \sin(\theta))^2 (a \cos(\theta)) (a \cos(\theta)) \, d\theta$$

$$= a^4 \int \sin^2(\theta) \cos^2(\theta) d\theta$$

Apply the double-angle identity and square

$$\left(\frac{1}{2} \sin(2\theta)\right)^2 = (\sin(\theta) \cos(\theta))^2$$

$$a^4 \int \frac{1}{4} \sin^2(2\theta) d\theta$$

$$\frac{a^4}{4} \int \sin^2(2\theta) d\theta$$

Use the half-angle identity

$$\sin^2(u) = \frac{1 - \cos(2u)}{2}$$

Apply u -substitution

$$u = 2\theta \quad \sin^2(2\theta) = \frac{1 - \cos(4\theta)}{2}$$

$$\frac{a^4}{8} \int 1 - \cos(4\theta) d\theta$$

Convert the bounds $x = a \sin(\theta)$

At $x = 0$

$$0 = a \sin(\theta) \Rightarrow \theta = 0$$

At $x = a$

$$a = a \sin(\theta) \Rightarrow \theta = \frac{\pi}{2}$$

Integrate

$$\frac{a^4}{8} \left[\theta - \frac{1}{4} \sin(4\theta) \right]_0^{\frac{\pi}{2}}$$

Evaluate the bounds

$$\frac{a^4}{8} \left(\frac{\pi}{2} - \frac{1}{4} \sin\left(\frac{4\pi}{2}\right) \right) = \frac{a^4}{8} \left(\frac{\pi}{2} - \frac{1}{4} \sin(0) \right) = \frac{\pi a^4}{16}$$

$$\frac{a^4}{8} \left(0 - \frac{1}{4} \sin(4 \cdot 0) \right) = 0$$

Finally

$$\boxed{\frac{1}{16} \pi a^4}$$
